Lecture 1 (Introduction and Review of Linear Algebra)

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The syllabus can be found on Canvas.

Linear Optimization

In calculus, we can find optimal solutions by taking the derivative of our objective function and setting it to 0. Afterwards, we can solve to determine which inputs lead to the maximum or minimum value of the function within the region we are searching.

Ex 1)

Consider the simple linear function

$$y = 4x + 1$$

If we wish to find the minimum value of y when $x \in [-3, 3]$, we first take the derivative

$$y' = 4$$

as the derivative is non-zero, there cannot be a minimum value on $(-3, 3)$. Now, we need to check the end points.

$$y(-3) = 4(-3) + 1 = -11$$

$$y(3) = 4(3) + 1 = 13$$

Notice that -11 is the smallest value of all of the candidates, so $y = 11$ is the minimum value of y on the domain and it is minimized at $x = -3$.

Ex 2)

Consider the function

$$f(x, y) = x + y$$

subject to $x, y \in [0, 1]$. Suppose we wish to maximize f. Then we take the gradient

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) = (1, 1)$$

As the gradient is constant and non-zero, then the minimal value cannot lie inside the square. It needs to lie on the boundary of the square. It turns out the maximum occurs at $(1, 1)$ and this maximum is $f(1, 1) = 2$.

Definition

It turns out that for any linear objective with a set of linear constraints that an optimal value will always lie on a vertex in the boundary of the feasible region. This is what we will spend most of our class learning. Let's start with some definitions. A **linear optimization problem (also called a linear program)** is an optimization problem of the following form:

$$\text{maximize (minimize) } f(x_1, x_2, \ldots, x_n) = c_0 + c_1 x_1 + \ldots c_n x_n$$

subject to (s.t.)

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \leq b_1$$

$$a_{21}x_1 + a_{22}x_2 \cdots + a_{2n}x_n \leq b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \leq b_m$$

Where $c_i, a_{ij}, b_j \in \mathbb{R}$ for $1 \leq i \leq n, 1 \leq j \leq m$ are arbitrary constants. In practice, this is a lot to write, so we usually formulate linear programs in terms of matrices and vectors.

$$\text{maximize (minimize) } f(\vec{x}) = c_0 + \vec{c} \cdot \vec{x}$$

s.t.

$$A\vec{x} \leq \vec{b}$$

where

$$\vec{c} = \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}, \ \vec{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \ \vec{b} = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}, \ A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}$$

and where the inequality in $A\vec{x} \leq \vec{b}$ is applied elementwise (Each row of $A\vec{x}$ is $\leq$ the corresponding row in $\vec{b}$). Because we can formulate linear programs in terms of matrices and vectors, we can use tools from linear algebra when studying them.

Linear Algebra Review

Linear Systems

A **linear system of equations** is a system of equations in the form:

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 \cdots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m$$

Where $a_{ij}, b_j \in \mathbb{R}$. This can be written more concisely in terms of matrices and vectors.

$$A\vec{x} = \vec{b}$$

Where

$$\vec{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \ \vec{b} = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}, \ A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}$$

If we row reduce the augmented matrix $A|b$, we can determine if the system of equations has a solution and if this solution is unique. If the row reduced matrix has a a leading 1 in each column, then the system of equations has a unique solution. If there is a row of all 0s in A after row reducing, then the system of equations will be inconsistent if the corresponding entry in the rightmost column of $A|b$ is non-zero. If this entry is 0 for each row of all 0s, then the system has infinitely many solutions.

Ex 1)

Consider the system

$$x + y = 0$$

$$x - y = 2$$

This yields the matrix formulation

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$

We get the augmented matrix

$$A|b = \left(\begin{array}{cc|c} 1 & 1 & 0 \\ 1 & -1 & 2 \end{array}\right) \xrightarrow{R_2 - R_1} \left(\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & -2 & 2 \end{array}\right) \xrightarrow{-\frac{1}{2}R_2} \left(\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 1 & -1 \end{array}\right) \xrightarrow{R_1 - R_2} \left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & -1 \end{array}\right)$$

As each column in the row reduced matrix A has a leading 1, a unique solution exists. This solution is found in the right side of the row reduced augmented matrix. $x = 1, y = -1$

Ex 2)

$$x + y + z = 2$$

$$2y + z = 3$$

$$x - y = 1$$

This yields the matrix formulation

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$$

And therefore, the augmented matrix

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 2 & 1 & 3 \\ 1 & -1 & 0 & 1 \end{array}\right) \xrightarrow{R_3 - R_1} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 2 & 1 & 3 \\ 0 & -2 & -1 & -1 \end{array}\right) \xrightarrow{R_3 + R_2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 2 & 1 & 3 \\ 0 & 0 & 0 & 2 \end{array}\right)$$

As there is a row of all 0s in the bottom but the right hand side is non-zero, then the system of equations is inconsistent. No solution exists.

Ex 3)

$$x + y + z = 2$$

$$2y + z = 3$$

$$x - y = -1$$

This yields the matrix formulation

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}$$

And therefore, the augmented matrix

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 2 & 1 & 3 \\ 1 & -1 & 0 & -1 \end{array}\right) \xrightarrow{R_3 - R_1} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 2 & 1 & 3 \\ 0 & -2 & -1 & -3 \end{array}\right) \xrightarrow{R_3 + R_2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 2 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right) \xrightarrow{\frac{1}{2}R_2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & \frac{1}{2} & \frac{3}{2} \\ 0 & 0 & 0 & 0 \end{array}\right) \xrightarrow{R_1 - R_2} \left(\begin{array}{ccc|c} 1 & 0 & & \\ 0 & 1 & & \\ 0 & 0 & & \end{array}\right.$$

The matrix A is row reduced and all rows containing only 0s also contain a 0 in the rightmost column of the augmented matrix. As such, this system has infinitely many solutions.

Subspaces

Let V be a vector space. A nonempty subset $S \subseteq V$ is a vector space if it satisfies the following for all $c \in \mathbb{R}$ and $\vec{x}, \vec{y} \in S$:

1) $\vec{x} + \vec{y} \in S$

2) $c\vec{x} \in S$

Since this vector space is contained within another vector space, we call S a **subspace** of V. Let's give some examples and non-examples.

Ex 4)

Let $V = \mathbb{R}^2$ and let $S = \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} | 2x_1 + x_2 = 0 \right\}$. Then S is a subspace. To prove this, we show both axioms hold. Let $\vec{x}, \vec{y} \in S$ and $c \in \mathbb{R}$. Then

1)

Both $\vec{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \vec{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$ are in S, therefore $2x_1 + x_2 = 2y_1 + y_2 = 0$, thus

$$\vec{x} + \vec{y} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \end{pmatrix}$$

And we see

$$2(x_1 + y_1) + (x_2 + y_2) = 2x_1 + 2y_1 + x_2 + y_2 = (2x_1 + x_2) + (2y_1 + y_2) = 0 + 0 = 0$$

2)

As $\vec{x} \in S$, then $2x_1 + x_2 = 0$, thus

$$c\vec{x} = c \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} cx_1 \\ cx_2 \end{pmatrix}$$

and we see

$$2cx_1 + cx_2 = c(2x_1 + x_2) = c0 = 0$$

As both **1)** and **2)** hold, we see S is a subspace of $\mathbb{R}^2$.

Non Ex 5)

Let $V = \mathbb{R}^2$ and let $S = \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} | 2x_1 + x_2 = 1 \right\}$. Then S is not a subspace. We can break this in a few ways. First let's show **1)** doesn't hold.

1)

Consider $\vec{x} = \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix}, \vec{y} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. As $2\left(\frac{1}{2}\right) + 0 = 2(0) + 1 = 1$, then both $\vec{x}, \vec{y} \in S$. Even so, we see

$$\vec{x} + \vec{y} = \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ 1 \end{pmatrix}$$

But

$$2\left(\frac{1}{2}\right) + 1 = 2 \neq 1$$

so

$$\vec{x} + \vec{y} \notin S$$

2)

Consider $\vec{x} = \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix}$ and $c = 2$. As we showed earlier, $\vec{x} \in S$. Even so, we see

$$2\vec{x} = 2 \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

but

$$2(1) = 2 \neq 1$$

So

$$2\vec{x} \notin S$$

Showing any axiom fails in at least one case is enough to show S is not a subspace. Here we showed a couple of ways that the axioms fail for S.

Ex 6)

Let $T : V \to W$ be a linear transformation from vector space V into vector space W. Consider the space

$$S = \text{Null}\,(T) = \{\vec{v} | T\,(\vec{v}) = 0\}$$

Then S is a subspace (called the null space of T). We will show this by showing both axioms hold. Let $\vec{x}, \vec{y} \in S$ and $c \in \mathbb{R}$.
1)
As $\vec{x}, \vec{y} \in S$, then $T\,(\vec{x}) = T\,(\vec{y}) = 0$, thus

$$T\,(\vec{x} + \vec{y}) = T\,(\vec{x}) + T\,(\vec{y}) = 0 + 0 = 0$$

As T is a linear transformation.
2)
As $\vec{x} \in S$, then $T\,(\vec{x}) = 0$,

$$T\,(c\vec{x}) = cT\,(\vec{x}) = c0 = 0$$

Ex 7)

Let V be a vector space and suppose that S, T are both subspaces of V. Then $S \cap T$ (The collection of all vectors in both S and T) is also a subspace. Let $\vec{x}, \vec{y} \in S \cap T$ and $c \in \mathbb{R}$.
1)
As S, T are both subspaces of V, and $\vec{x}, \vec{y} \in S, T$ then $\vec{x} + \vec{y} \in S, T$ therefore $\vec{x} + \vec{y} \in S \cap T$
2)
As S, T both subspaces of V, and $\vec{x} \in S, T$ then $c\vec{x} \in S, T$ therefore $c\vec{x} \in S \cap T$

Eigenstuffs

Let A be an $n \times n$ matrix. A pair $(\lambda, \vec{v})$ are called an **eigenpair** (with λ as the **eigenvalue** and $\vec{v}$ as the **eigenvector**) if

$$A\vec{v} = \lambda\vec{v}$$

if $\vec{v}$ is nonzero. We can find eigenpairs using the determinant. Note the following:

$$A\vec{v} = \lambda\vec{v}$$

$$A\vec{v} - \lambda\vec{v} = 0$$
$$A\vec{v} - \lambda I\vec{v} = 0$$

$$(A - \lambda I)\,\vec{v} = 0$$

But as $\vec{v}$ is non-zero, then $A - \lambda I$ must be singular. We can find which values of λ yield a singular matrix by taking the determinant, setting it equal to 0, then solving for λ. For each root, λ, we then can solve the linear system formed by plugging in λ into $(A - \lambda I)\,\vec{v} = 0$.

Ex 8)

$$A = \begin{pmatrix} -2 & 1 \\ 3 & 0 \end{pmatrix}$$

If we wish to find the eigenstuffs, we start by finding the eigenvalues:

$$A - \lambda I = \begin{pmatrix} -2-\lambda & 1 \\ 3 & -\lambda \end{pmatrix} = -\lambda(-2-\lambda) - 3 = \lambda^2 + 2\lambda - 3 = 0$$

$$(\lambda + 3)(\lambda - 1) = 0$$

Thus $\lambda_1 = -3$, $\lambda_2 = 1$. Now, we can find the eigenvectors. First for λ_1,

$$A - \lambda_1 I = A + 3I = \begin{pmatrix} 1 & 1 \\ 3 & 3 \end{pmatrix}$$

So

$$(A - \lambda_1 I)\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 3 & 3 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0$$

So we can solve the linear system

$$\left(\begin{array}{cc|c} 1 & 1 & 0 \\ 3 & 3 & 0 \end{array}\right) \Longrightarrow \left(\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right)$$

So $x + y = 0$ so $y = -x$ thus $\vec{v}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$. Here, we set $x = 1$ arbitrarily as we only need to find a basis vector. Now, for $\lambda_2 = 1$:

$$A - \lambda_2 I = A - I = \begin{pmatrix} -3 & 1 \\ 3 & -1 \end{pmatrix}$$

So

$$(A - \lambda_2 I)\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -3 & 1 \\ 3 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0$$

So if we solve the linear system:

$$\left(\begin{array}{cc|c} -3 & 1 & 0 \\ 3 & -1 & 0 \end{array}\right) \Longrightarrow \left(\begin{array}{cc|c} -3 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right)$$

So $-3x + y = 0$ thus $y = 3x$. If we set $x = 1$, then we find $\vec{v}_2 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$.

Ex 9)

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{pmatrix}$$

First, we will find the eigenvalues.

$$A - \lambda I = \begin{pmatrix} 1-\lambda & 1 & 0 \\ 0 & 2-\lambda & 2 \\ 0 & 0 & 3-\lambda \end{pmatrix}$$

It can be verified

$$\det\begin{pmatrix} 1-\lambda & 1 & 0 \\ 0 & 2-\lambda & 2 \\ 0 & 0 & 3-\lambda \end{pmatrix} = (1-\lambda)(2-\lambda)(3-\lambda)$$

as this is an upper triangular matrix. I encourage you to verify this! As such, we get

$$(1-\lambda)(2-\lambda)(3-\lambda) = 0$$

So $\lambda_1 = 1,\ \lambda_2 = 2,\ \lambda_3 = 3$. Now, we can find the corresponding eigenvectors.

$$(A - \lambda_1 I)\,\vec{v}_1 = (A - I)\,\vec{v}_1 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

Thus

$$y = 0,\ y + 2z = 0,\ 2z = 0$$

Thus x is free, and $y, z = 0$. This means we get $\vec{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$. Now, for the second eigenvector:

$$(A - \lambda_2 I)\,\vec{v}_2 = \begin{pmatrix} 1-2 & 1 & 0 \\ 0 & 2-2 & 2 \\ 0 & 0 & 3-2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -1 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

Thus

$$-x + y = 0,\ 2z = 0,\ z = 0$$

So

$$x = y,\ z = 0$$

This yields the eigenvector $\vec{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$. Finally, for λ_3, we have

$$(A - \lambda_3 I)\,\vec{v}_3 = \begin{pmatrix} 1-3 & 1 & 0 \\ 0 & 2-3 & 2 \\ 0 & 0 & 3-3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -2 & 1 & 0 \\ 0 & -1 & 2 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

So we can row reduce to solve:

$$\left(\begin{array}{ccc|c} -2 & 1 & 0 & 0 \\ 0 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \Longrightarrow \left(\begin{array}{ccc|c} -2 & 0 & 2 & 0 \\ 0 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \Longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Thus

$$x - z = 0,\ y - 2z = 0$$

So

$$x = z,\ y = 2z$$

Yielding the eigenvector

$$\vec{v}_3 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

when $z = 1$.